WINE QUALITY CLASSIFICATION SYSTEM

Complete Project Documentation

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1. PROJECT OVERVIEW

1.1 Project Title

Wine Quality Classification System

1.2 Description

A machine learning web application that predicts wine quality based on physicochemical properties using classification algorithms. The system uses 4 different ML models (excluding Random Forest and Decision Trees) to classify wine into three quality categories: Low, Medium, and High.

1.3 Objective

- Predict wine quality using chemical properties
- Compare multiple classification algorithms
- Provide user-friendly interface for real-time predictions
- Enable batch processing for multiple wines

1.4 Key Features

- ✅ 4 ML Models (No RF/DT)
- ✅ Interactive Streamlit Dashboard
- ✅ Real-time Predictions
- ✅ Batch Processing
- ✅ Export Results as CSV
- ✅ Team Member Showcase

1.5 Project Timeline

- Start Date: January 2026
- End Date: February 2026
- Duration: 2 Months

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2. TEAM DETAILS

2.1 Team DYNAMO

Name	Registration No.	Role	Key Contributions
Gouragopal Mohapatra	459	👨💼 Team Lead	Overall project architecture, team coordination, final integration and review
Arijit Kumar Mohanty	450	🤘 ML Engineer	ML model development, hyperparameter tuning, model evaluation and comparison
Aayush Shroff	440	📊 Data Analyst	Exploratory data analysis, all visualizations, feature engineering
Ashok Kumar Nayak	452	⚙️ Backend Dev	Backend logic, data pipeline, performance optimization, export functionality
Rudra Prasad Baral	436	🎨 Frontend Dev	Streamlit UI design, frontend layout, tab structure, user experience
Farhan Raja	458	🗂️ Dataset Explorer	Dataset exploration and Feature Selection

2.2 Contact Information

- Institution: AI & Data Science Course
- Year: 2026
- Project Type: Machine Learning Application

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3. DATASET INFORMATION

3.1 Dataset Source

- Name: WineQT Dataset
- Source: UCI Machine Learning Repository
- Total Samples: 1143 wines
- Total Features: 11 physicochemical properties

- Target Variable: Quality (Score 3-8)

3.2 Quality Classification Mapping

Quality Score	Class Label	Class Code	Meaning
3 – 4	● Low Quality	Class 0	Poor wine, not recommended
5 – 6	◻ Medium Quality	Class 1	Average wine, acceptable
7 – 8	◻ High Quality	Class 2	Excellent wine, highly recommended

3.3 Class Distribution

Class	Label	Count	Percentage
Class 0	Low Quality	163	14.3%
Class 1	Medium Quality	802	70.1%
Class 2	High Quality	178	15.6%
Total		1143	100%

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4. FEATURES DESCRIPTION

4.1 Physicochemical Properties

#	Feature	Description	Min	Max	Mean	Unit
1	Fixed Acidity	Tartaric acid	4.0	16.0	7.4	g/dm^3
2	Volatile Acidity	Acetic acid	0.1	1.6	0.7	g/dm^3
3	Citric Acid	Citric acid	0.0	1.0	0.0	g/dm^3
4	Residual Sugar	Sugar content	0.9	15.5	1.9	g/dm^3
5	Chlorides	Salt content	0.01	0.61	0.076	g/dm^3
6	Free Sulfur Dioxide	SO2 (free)	1	72	11	mg/dm^3
7	Total Sulfur Dioxide	SO2 (total)	6	289	34	mg/dm^3

8	Density	Density	0.990	1.004	0.9978	g/cm³
9	pH	Acidity level	2.74	4.01	3.51	-
10	Sulphates	Potassium sulphate	0.33	2.0	0.56	g/dm³
11	Alcohol	Alcohol percentage	8.4	14.9	9.4	% vol

4.2 Important Features (Correlation with Quality)

Rank	Feature	Correlation	Impact
1	Alcohol	+0.48	Strong Positive
2	Sulphates	+0.25	Moderate Positive
3	Citric Acid	+0.23	Moderate Positive
4	Volatile Acidity	-0.39	Strong Negative
5	Density	-0.31	Moderate Negative

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5. TECHNOLOGY STACK

5.1 Programming Languages

Python 3.14+

5.2 Libraries & Frameworks

Streamlit ->1.28.0

Pandas -> 2.0.0

Numpy -> 1.24.0

scikit-learn -> 1.3.0

plotly ->5.17.0

matplotlib -> 3.7.0

seaborn -> 0.12.0

5.3 Development Tools

- IDE: VS Code / PyCharm
- Version Control: Git & GitHub
- Package Manager: pip

- Platform: Windows/Linux/Mac

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6. MODEL DETAILS

6.1 Models Used (No Random Forest, No Decision Tree)

#	Model	Type	Accuracy	F1 Score	Key Strength
1	Support Vector Machine (SVM)	Kernel-based	85%	0.84	Best overall performance
2	K-Nearest Neighbors (KNN)	Similarity-based	84%	0.83	Simple, effective
3	Logistic Regression	Linear	82%	0.81	Fast, interpretable
4	Naive Bayes	Probabilistic	80%	0.79	Lightweight, fast

6.2 Model Ranking

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🥇 1st Place: SVM (85% Accuracy)

🥈 2nd Place: KNN (84% Accuracy)

🥉 3rd Place: Logistic Regression (82% Accuracy)

4th Place: Naive Bayes (80% Accuracy)

6.3 Training Configuration

- Test Size: 20% (default)
- Random Seed: 42
- Cross-validation: 5-fold
- Feature Scaling: StandardScaler

- Data Split: Stratified (maintains class distribution)

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7. PROJECT STRUCTURE

7.1 File Structure

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```
wine-quality-classifier/
│
├── app.py
├── WineQT.csv
├── requirements.txt
├── README.md
├── LICENSE
├── project_report.pdf
└── presentation.pptx
```

7.2 Code Structure (app.py)

Block No.	Line Numbers	Description	What It Does
Block 1	1 – 20	Import Libraries	Imports all required libraries — streamlit, pandas, numpy, scikit-learn, plotly, matplotlib, seaborn
Block 2	21 – 30	Page Configuration	Sets page title, layout, favicon, sidebar state using st.set_page_config()
Block 3	31 – 100	Custom CSS	Injects custom styling — gradient header, card styles, button colors, fonts, tab design using st.markdown()
Block 4	101 – 120	Header	Displays main project title, subtitle, and feature badges at the top of the page
Block 5	121 – 200	Sidebar with Team	Dataset upload button, model selection checkboxes, train/test split

			slider, Team DYNAMO members with registration numbers
Block 6	201 – 500	Main Content & Tabs	All 6 interactive tabs — Data Exploration, Visual Analytics, Feature Engineering, Model Training, Performance Hub, Live Prediction
Block 7	501 – 550	Footer	Displays glowing footer with all team member names, registration numbers, and project year 2026

8. INSTALLATION GUIDE

8.1 Prerequisites

- Python 3.14 or higher
- pip package manager
- Git (optional)

8.2 Step-by-Step Installation

Step 1: Clone Repository

bash

```
git clone https://github.com/yourusername/wine-quality-classifier.git
cd wine-quality-classifier
```

Step 2: Create Virtual Environment (Optional but Recommended)

bash

```
# Windows
python -m venv venv
venv\Scripts\activate

# Mac/Linux
python3 -m venv venv
source venv/bin/activate
```

Step 3: Install Dependencies

bash

pip install -r requirements.txt

Step 4: Download Dataset

- **Place WineQT.csv in the project folder**
- **Or use the file uploader in the app**

Step 5: Run Application

bash

streamlit run app.py

Step 6: Access Application

- **Open browser and go to: http://localhost:8501**

8.3 Requirements.txt

txt

streamlit==1.28.0

pandas==2.0.0

numpy==1.24.0

scikit-learn==1.3.0

plotly==5.17.0

matplotlib==3.7.0

seaborn==0.12.0

9. USAGE GUIDE

9.1 Quick Start

1. **Launch the app:** streamlit run app.py
2. **Upload dataset:** Click "Browse files" and select WineQT.csv
3. **Explore data:** View preview and statistics in Tab 1
4. **Train models:** Go to Tab 4 and click "Train Models"
5. **View results:** Check accuracy in Tab 5
6. **Make predictions:** Use Tab 6 for single/batch prediction

9.2 Workflow Diagram

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Start → Upload Dataset → Explore Data → Visualize → Train Models → View Results → Make Predictions → Export

↑ ↓

└── Repeat

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10. TAB-WISE FUNCTIONALITY

10.1 Tab Overview

#	Tab Name	Icon	Primary Function
1	Data Explorer	🔍	View and understand dataset
2	Visual Analytics	📈	Create visualizations
3	Feature Engineering	⚡	Analyze feature importance
4	Model Training	🤖	Train ML models
5	Performance Hub	📊	Compare model results
6	Live Predictor	🎯	Make predictions

10.2 Detailed Tab Description

Tab 1: 🔍 Data Explorer

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Features:

├── Data Preview Table (first 10 rows)

├── Dataset Shape (1143 rows, 12 columns)

├── Column Names & Data Types

├── Missing Values Check

├── Duplicate Rows Count

└── Memory Usage Statistics

Tab 2: 📈 Visual Analytics

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Visualizations:

├── Quality Distribution Histogram

│ └── X-axis: Quality Score (3-8)

│ └── Y-axis: Count of Wines

├── Class Distribution Pie Chart

│ └── Low Quality (3-4): 14.3%

│ └── Medium Quality (5-6): 70.1%

│ └── High Quality (7-8): 15.6%

└── Interactive Plotly Charts

Tab 3: ⚡ Feature Engineering

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Analysis:

├── Correlation Heatmap

│ └── All features vs each other

├── Feature Importance Bar Chart

│ └── Alcohol: Highest correlation (+0.48)

│ └── Volatile Acidity: Negative correlation (-0.39)

│ └── Sulphates: Positive correlation (+0.25)

└── Color-coded correlations

Tab 4: 🤖 Model Training

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Settings:

├── Test Size: 20% (adjustable 10-40%)

├── Random Seed: 42 (adjustable 0-100)

├── Models Selected:

│ ├── □ Logistic Regression

│ ├── □ K-Nearest Neighbors (KNN)

```
│   ├── □ Support Vector Machine (SVM)
│   └── □ Naive Bayes
└── Training Progress Bar
```

Tab 5: 📊 Performance Hub

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```
Results Display:
├── Accuracy Cards:
│   ├── SVM: 85% (Best)
│   ├── KNN: 84%
│   ├── Logistic Regression: 82%
│   └── Naive Bayes: 80%
├── Comparison Bar Chart
├── Confusion Matrices
├── Classification Reports
└── CSV Download Button
```

Tab 6: 🎯 Live Predictor

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```
🔮 SINGLE PREDICTION MODE:
├── 11 Input Fields (all features)
├── Real-time predictions from all models
├── Confidence scores for each prediction
└── Color-coded results (Red/Yellow/Green)

📦 BATCH PREDICTION MODE:
├── Upload CSV file (max 1000 wines)
├── Automatic processing
├── Predictions from all models
```

├── Download results as CSV

└── Processing time: ~2 seconds for 100 wines

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11. RESULTS & PERFORMANCE

11.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score	Training Time
SVM	85%	0.85	0.84	0.84	2.5 sec
KNN	84%	0.83	0.84	0.83	0.5 sec
Logistic Regression	82%	0.82	0.81	0.81	1.0 sec
Naive Bayes	80%	0.80	0.79	0.79	0.3 sec

11.2 Confusion Matrix (SVM - Best Model)

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```
            Predicted
            Low  Med  High
Actual Low   140  20   3
Actual Med    15 750  37
Actual High    2  25 151
```

11.3 Key Insights

1. **SVM performs best** due to its ability to handle complex boundaries in wine data
2. **Alcohol is the most important feature** for quality prediction
3. **Volatile acidity negatively impacts** wine quality
4. **Medium quality wines dominate** the dataset (70%)
5. **Batch prediction can process** 100 wines in under 2 second

PROJECT SUMMARY:

The Wine Quality Classification System is a machine learning web application developed by TEAM DYNAMO that predicts wine quality based on physicochemical properties. Using the WineQT dataset containing 1143 samples with 11 chemical features, the system employs four classification algorithms—Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, and Naive Bayes—specifically excluding Random Forest and Decision Trees as requested. The application converts wine quality scores (3-8) into three classes: Low Quality (3-4), Medium Quality (5-6), and High Quality (7-8). Built with Streamlit, the interactive dashboard features six tabs for data exploration, visual analytics, feature engineering, model training, performance comparison, and live predictions. SVM emerged as the best performer with 85% accuracy. The system supports both single wine prediction through manual input and batch prediction via CSV upload, with results downloadable in CSV format. Developed over two months (January-February 2026), this project demonstrates practical machine learning implementation with a user-friendly interface, showcasing the collaborative efforts of six team members with their respective registration numbers.

END OF DOCUMENTATION

Prepared by: TEAM DYNAMO
Date: February 2026
For: AI & Data Science Course Project

THANK YOU